When does a river forecast change a decision?

Flood warning at Gandak Barrage and water allocation on the Potomac

Abstract

Could a river forecast change a decision in time? We replayed two decisions using historical inputs and checked the outcomes afterwards. On the Gandak River, a model using barrage inflow, upstream water levels and forecast rain signalled six hours before the barrage crossed a study-defined high-flow threshold in September 2026. Random forest and gradient boosting, tested on identical inputs, signalled at the same hour as ridge regression. Khadda had already reached its official warning level one hour earlier. The signal preceded the higher danger level by 47 hours. Village loudspeaker warnings were documented, but their first receipt time is unknown. Only one hourly flood episode could be tested. On the Potomac, we forecast next-day water use from the calendar and compared it with reconstructed municipal withdrawals on 2,041 days. Mean absolute demand error was 106,378 m³/day, 34% lower than using the historical average. With that demand forecast, planning against a conservative river-flow estimate cut overcommitment from 36.6% to 1.0% of days compared with the ensemble mean, while leaving an extra 98,699 m³/day of demand unmet. The replays show how the location of a gauge and the amount a planner is willing to promise change the practical value of a forecast.